

Chapter 7 pt 1: Reading and Writing Files

EEC 3150

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Outline

- 1 Basic of Reading Files
- 2 Extracting Data from Files

Basic of Reading Files

Basic of Reading Files

Intro to reading a file with Python:

```
fname = "FirstFile.txt"

fhandle = open(fname, 'r')

string = fhandle.read()
print(string)

fhandle.close()
```

- To read a file, use the `open()` function
- Two string arguments: 1) the filename, 2) whether to read ('r'), write ('w'), or append ('a') to the file
- Returns a *file handle* (essentially a variable that points to the file)
- Since file handles are objects, they have their own methods (such as `read()`)
- Close the file when finished
- Trying to read after closing throws an error

Basic of Reading Files

Reading files in a with,as block:

- Closing a file is annoying and you can forget to do it
- For convenience, we can read files inside a with,as block

```
fname = "FirstFile.txt"
with open(fname, 'r') as fhandle:
    string = fhandle.read()
    print(string)
```

- The file handle is only defined in the scope of the block
- Variables you define inside the block will still be defined after the block
- The file is closed automatically after the block is finished
- Read the file in the block, save the contents to a variable, and use them later on

Basic of Reading Files

Some read methods:

```
with open(fname, 'r') as fhandle:  
    string = fhandle.read()
```

`fhandle.read()`

- Returns the entire file in one string

```
with open(fname, 'r') as fhandle:  
    lines = fhandle.readlines()
```

`fhandle.readlines()`

- Returns a list of strings split on newlines

Basic of Reading Files

Reading a line at a time:

```
with open(fname, 'r') as fhandle:  
    for line in fhandle:  
        print(line)
```

```
with open(fname, 'r') as fhandle:  
    line = fhandle.readline()  
    print(line)  
    line = fhandle.readline()  
    print(line)
```

With a for loop:

- We can read a line at a time with a for loop

`fhandle.readline()`

- Returns the next line

Basic of Reading Files

Reading a line at a time:

```
with open(fname, 'r') as fhandle:
    line = fhandle.readline()
    while line:
        print(line)
        line = fhandle.readline()
```

With a while loop:

- We can read a line at a time with a while loop
- Read the first line
- Use the line as the loop condition
- Once `readline()` returns an empty string, the loop exits

Extracting Data from Files

Extracting Data from Files

Get a list of grades from a file:

Read the file Grades.txt and produce a list of integer grades.

- This file is simple and has one number per line

Get lists of years and temperatures from a file:

Read the file YearTemp.txt and produce lists of integer years and float temperatures.

- This file is more complicated
- It has a header that we must ignore
- Year and temp values are comma-separated (one pair per line)

Extracting Data from Files

Get lists of years and temperatures from a file:

Read the file YearTemp.txt and produce lists of integer years and float temperatures.

- This file is more complicated
- It has a header that we must ignore
- Year and temp values are comma-separated (one pair per line)